

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 9 - Formative Assessment 4

Formative Assessment (FA-4)

Class: Class 9

Assessment Type: Formative Assessment (FA-4)

Total Marks: 30

Duration: 90 minutes

Topics Covered: Web Technologies - HTML, CSS, JavaScript

ICT Teaching Resources Portal

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Part A: Written Concept Paper (20 Marks)

Section I: MCQ (5 x 1 = 5 marks)

- a. Which HTML5 tag is used to define a navigation section?
 - b. <menu>
 - c. <nav>
 - d. <navigation>
 - e. <links>
- b. Which CSS property is used to create a flexible layout?
 - c. display: block
 - d. display: inline
 - e. display: flex
 - f. display: table
- c. What does `document.getElementById("demo").innerHTML = "Hello";` do?
 - d. Changes the CSS of element
 - e. Changes the HTML content of element
 - f. Deletes the element
 - g. Creates a new element
- d. Which event occurs when the user clicks on an HTML element?
 - e. onmouseover
 - f. onchange
 - g. onclick
 - h. onmouseclick
- e. Which CSS unit is relative to the font-size of the root element?
 - f. em
 - g. rem
 - h. px
 - i. %

Section II: Short Answer (5 x 2 = 10 marks)

- a. Differentiate between **inline CSS**, **internal CSS**, and **external CSS**. When would you use each?
- b. Write the HTML and CSS code to create a responsive navigation bar using flexbox that collapses into a hamburger menu on mobile screens (write the structure only).
- c. Explain the CSS **box model** with a labelled diagram. How does box-sizing: border-box change the default behaviour?
- d. Write JavaScript code to:
- e. Validate that a form field is not empty
- f. Display an alert with the field value if valid
- g. Show an error message if empty
- e. What is the difference between var, let, and const in JavaScript? Give one example of each.

Section III: Long Answer (5 x 1 = 5 marks)

- a. Create a complete web page for a **Student Management System** that includes:
- b. An HTML form with fields: Name, Class, Section, Roll Number, and three subject marks
- c. CSS styling with a professional colour scheme, styled form inputs, and a submit button
- d. JavaScript to:
 - Calculate the total and percentage when the user clicks a “Calculate” button
 - Assign a grade based on the percentage
 - Display the results in a styled card below the form
 - Validate all fields before submission

Part B: Hands-on Practical Lab Task (5 marks)

Create a **Quiz Application** web page that:

- Displays one question at a time with 4 multiple-choice options
- Uses JavaScript to track the current question and score
- Highlights correct/incorrect answers with colour changes
- Shows a progress bar indicating current question out of total
- Displays the final score with a message at the end
- Uses CSS for attractive styling with animations

Part C: Notebook Maintenance (5 marks)

- a. Neatness and organisation of notebook (2 marks)
- b. Completion of all assigned classwork and homework (2 marks)
- c. Proper headings, dates, and indexing (1 mark)